MASK THE NAME.
NOT THE BIAS.

A dual audit for gender bias in ATS ranking — behavior and activation space.

87% of the score gap survives name masking.

The direction varies. The harm does not.

Gender is not a stable ‘female penalty’. It is a model- and job-specific signal — and it survives anonymization.



https://www.bbc.co.uk/news/technology-45809919

NEWS

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Technology

Amazon scrapped 'sexist AI' tool

10 October 2018

An algorithm that was being tested as a recruitment tool by online giant Amazon was sexist and had to be scrapped, according to a Reuters report.

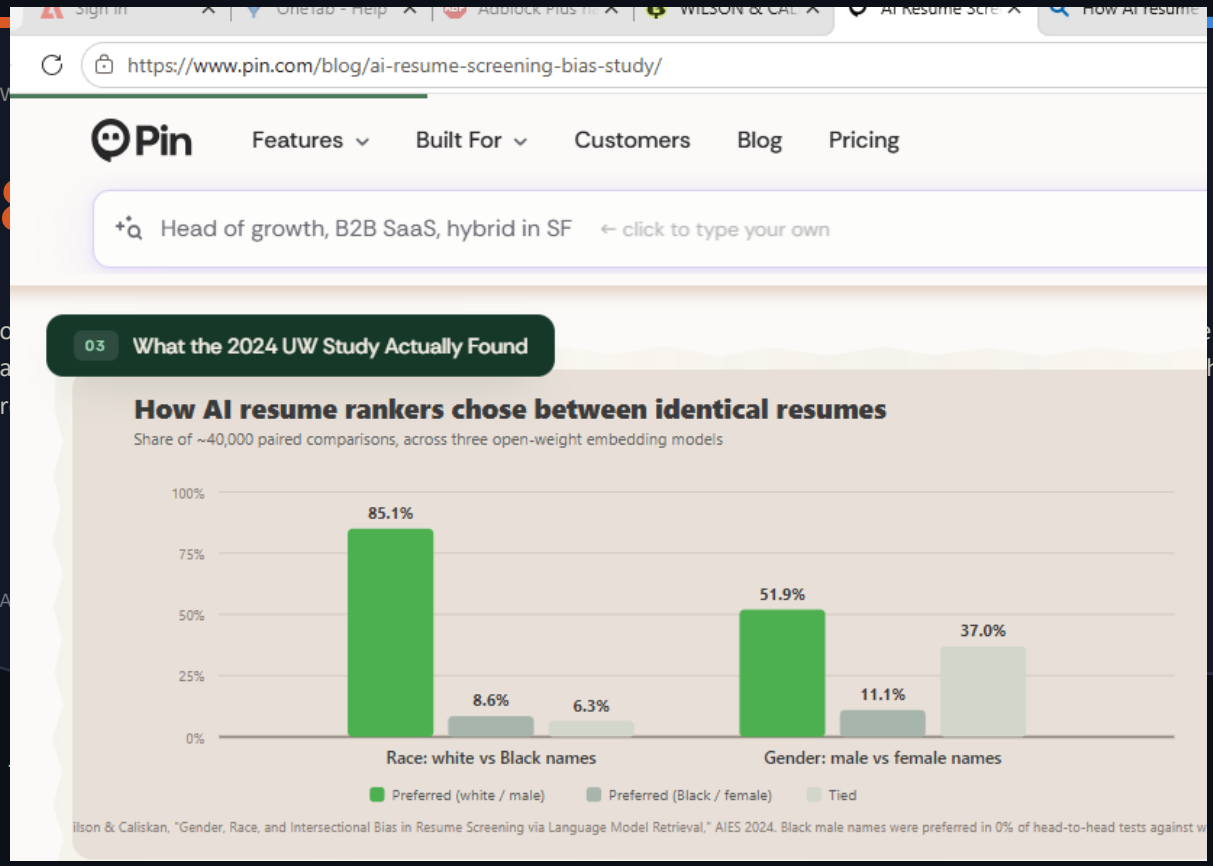
The artificial intelligence system was trained on data submitted by applicants over a 10-year period, much of which came from men, it claimed.

Reuters was told by members of the team working on it that the system effectively taught itself that male candidates were preferable.

Amazon has not responded to the claims.

Reuters spoke to five members of the team who developed the machine learning tool in 2014, none of whom wanted to be publicly named.

They told Reuters that the system was intended to review job applications and



https://www.pin.com/blog/ai-resume-screening-bias-study/

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03 What the 2024 UW Study Actually Found

How AI resume rankers chose between identical resumes

Share of ~40,000 paired comparisons, across three open-weight embedding models

ilson & Caliskan, "Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval," AIES 2024. Black male names were preferred in 0% of head-to-head tests against w

JobFair (2024) finds bias against males in at least one industry in 7/10 LLMs

SOURCES IN APPENDIX · AIES = AAAI/ACM Conference on AI, Ethics, and Society

Blind screening hides the name.

The model still sees gender.

Name masking removes one input. It does not remove the proxy cues that encode gender — or the direction those cues leave inside the model.

8.8 pp

score gap, full CV

female − male, matched pairs

100%

linear gender probe

after the name is masked

0.2 pp

residual gap after LEACE

same ranker, erased representation

PROTOTYPE BENCHMARK · 320 HELD-OUT CVS · NEGATIVE GAP = FEMALE CANDIDATES SCORE LOWER

#2

Sarah Nguyen

she/her · Bristol, UK

Senior Sales Manager · 5 years

MSc Public Health · CRM, negotiation, pipeline

8 projects · 27% KPI improvement

AFFILIATION

Girls Who Code mentor

PREDICTED SHORTLIST PROBABILITY

51.8%

−8.8 pp

#1

Michael Nguyen

he/him · Bristol, UK

Senior Sales Manager · 5 years

MSc Public Health · CRM, negotiation, pipeline

8 projects · 27% KPI improvement

AFFILIATION

Sigma Chi fraternity

PREDICTED SHORTLIST PROBABILITY

60.6%

Same qualifications. Different identity cues. The ranker moves the male candidate ahead.

Scores are real outputs from the prototype ranker; names and layout are a mock ATS view.

One cue masked. Five still leak.

A name is a proxy for gender. So are the words and affiliations around it.

THE CV / PARSED FIELDS

Name	MASKED
Pronouns	REMAINS
Affiliations	REMAINS
Career-gap reason	REMAINS
Gendered language	REMAINS
Career patterns	REMAINS



WHAT THE ENCODER SEES

The name is gone. The hidden state still separates genders.

87%

of the score gap survives name masking

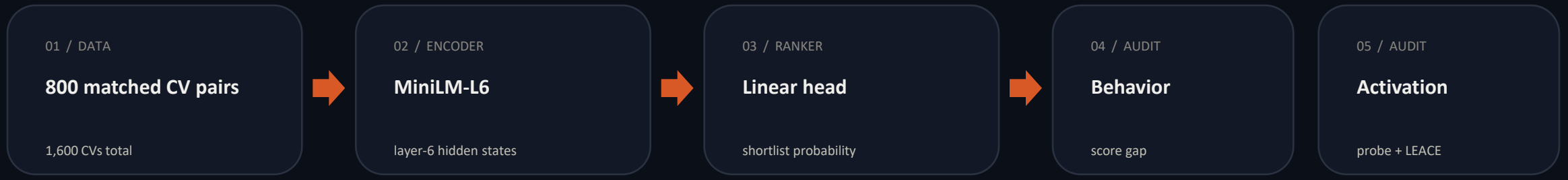
PROBE 100% · CHANCE 50%

The output looks blind. The representation is not.

Masking is a string operation. Bias is a representation operation.

A controlled benchmark with known ground truth.

We inject a known name-linked penalty, then test whether the audit recovers it — and removes it.



DESIGN CHOICES

8 job families · 10 features
80/20 split by candidate · no pair leakage
5-fold group CV for probes
synthetic labels with a known injected gap

WHAT IT BUYS US

A known bias direction, so we can measure
what the audit catches and what it misses.
The absolute numbers are not a production claim;
the causal comparison is the point.

Synthetic ground truth → measurable recovery → a test we can rerun on real logs.

10 features. Only 4 carry identity.

The ranker sees all 10. A fair ranker should be sensitive to the 6 merit features — and invariant to the 4 identity features.

IDENTITY-LINKED / 4

01

Name

Emily Carter
James Carter

02

Pronouns

she/her
he/him

03

Affiliation

women’s club
fraternity

04

Career-gap reason

parental leave
sabbatical

MERIT / 6

05

Skills match

4 of 6 required skills

06

Years experience

0–15 years

07

Education

BSc / MSc / PhD

08

Seniority

Junior → Lead

09

Certifications

0–4

10

Projects

1–8 shipped

Feature audit: name 42% of the gap · pronouns 31% · affiliation 5% · career-gap reason 0%

Paired name-swap audit.

Same qualifications. Swap only the identity cues. Compare scores and shortlist rates.

PROTOCOL

1 Match on merit features.

2 Swap name, pronouns, affiliation, gap reason.

3 Score both with the same ranker.

4 Compare paired differences, not group means.

Repeat across 8 job families.

SCORE GAP / PAIR i, JOB j

$$d_{ij} = s^{f}_{ij} - s^{m}_{ij}$$

The same CV, scored twice.

SHORTLIST GAP

$$P(\text{shortlist} \mid F) - P(\text{shortlist} \mid M)$$

At the real cutoff, not a proxy.

DEMOGRAPHIC PARITY

$$\Delta = |P(\hat{Y}=1 \mid A=F) - P(\hat{Y}=1 \mid A=M)|$$

A one-number selection check.

BALANCED ACCURACY

½ (qualified acceptance + unqualified rejection)

Catches a model that rejects everyone.

PASS GATE |score gap| ≤ 1 pp · |shortlist gap| ≤ 2 pp · AUC within 1 point of baseline

Behavioral bias is the distance between outcome distributions — not a single model card score.

Can a linear probe still read gender?

If a simple classifier recovers gender from the hidden state, the representation still carries it.

THE QUESTION

Does the model's internal state encode gender — after the name is gone?

READ-OUT

$P(\text{gender} \mid h_l)$

Chance = 50%. Above chance = linear information remains.

LINEAR PROBE

Logistic classifier on h_l

5-fold, grouped by CV pair

BIAS VECTOR

$v_l = E[h_l \mid F] - E[h_l \mid M]$

Mean difference direction

RELATIVE MAGNITUDE

$\|v_l\| / E\|h_l\|$

Scale-free signal size

PRESENCE

$\|v_l\| / \|\Sigma_l\|$

Bias relative to feature spread

RETENTION

R = (Presence(M') − Presence(B)) / (Presence(M_base) − Presence(B))

How much bias is left vs a blind baseline

COSINE SIMILARITY

cos(ψ_after, ψ_bias)

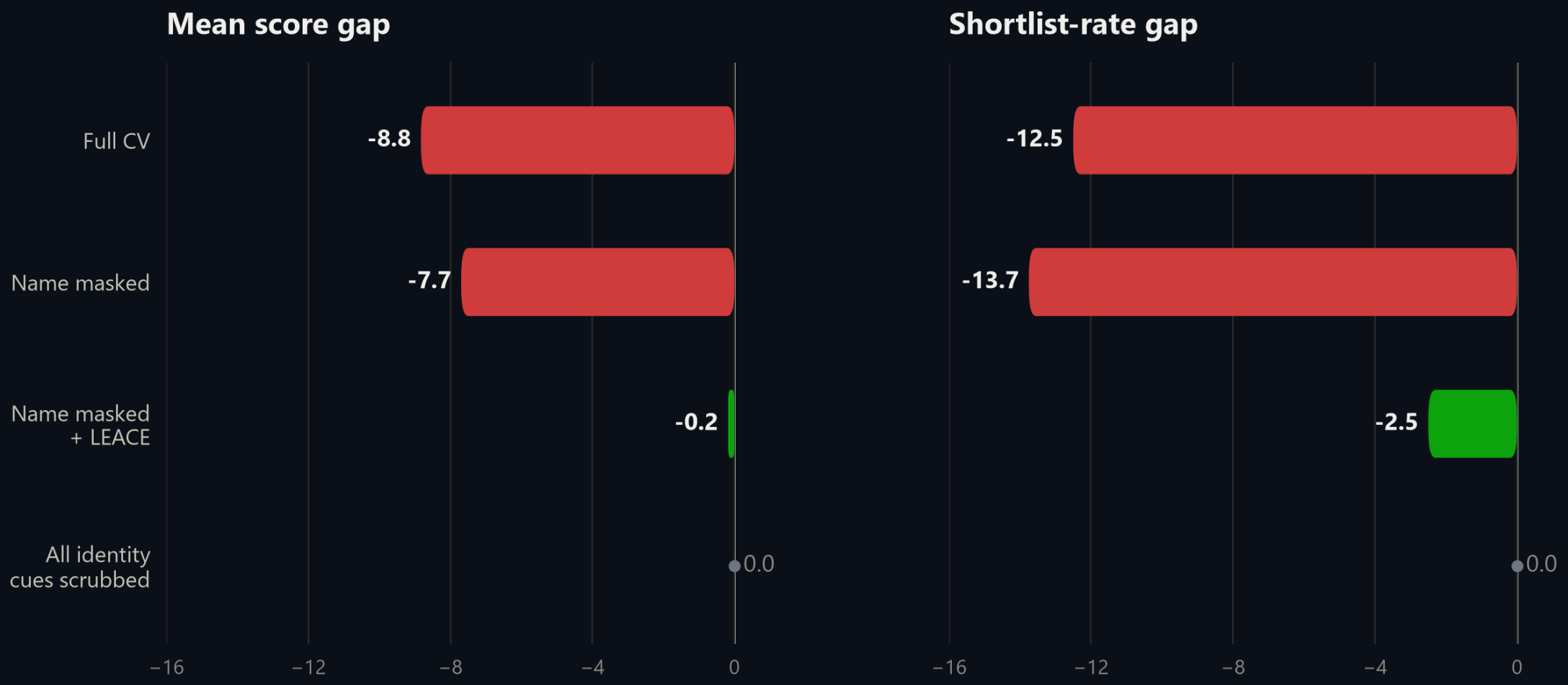
Alignment with the original direction

Behavior can look fair while the representation still separates groups.

All metrics computed at layer 6; probe accuracy reported out-of-fold.

The gap survives anonymization

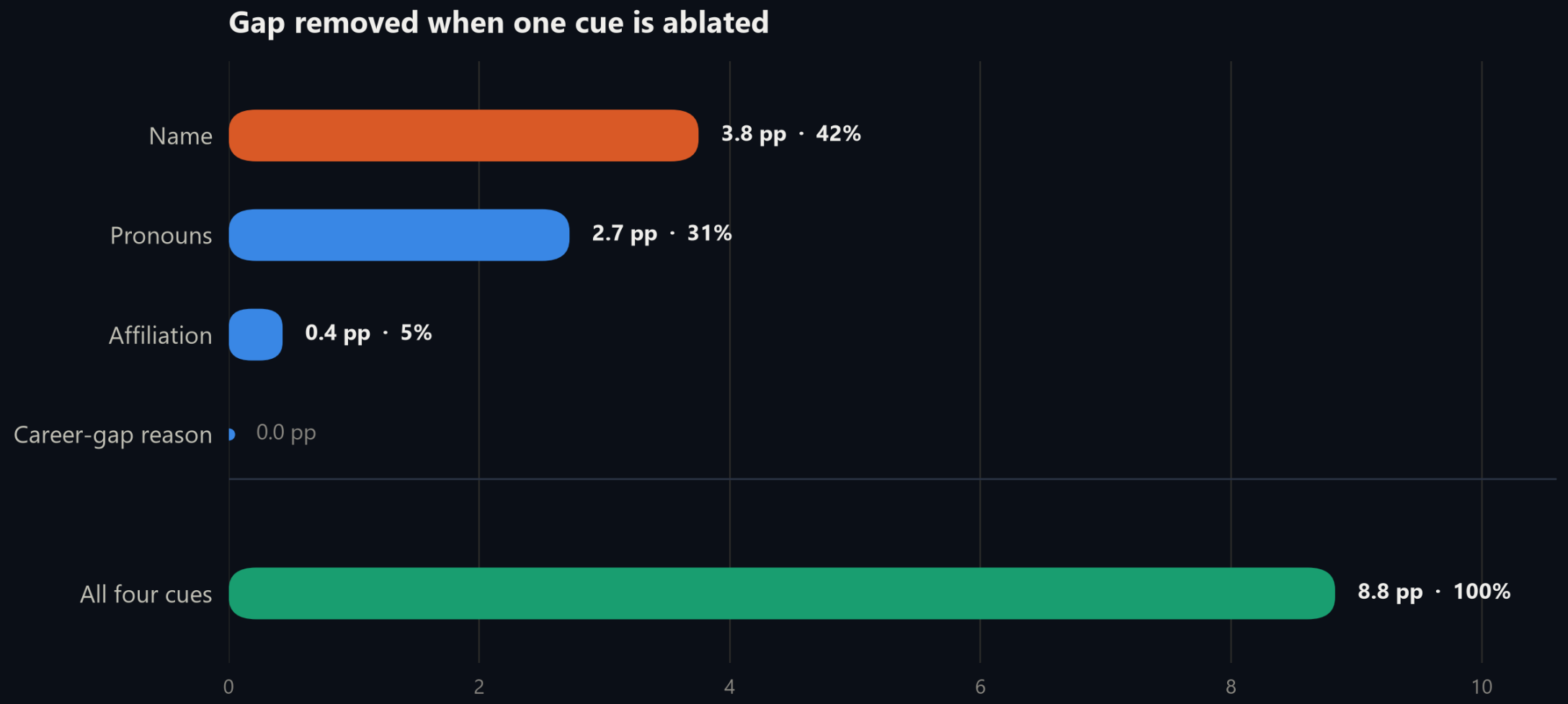
Female − male, percentage points. Held-out pairs, n = 320 CVs.



Negative = female candidates score lower. 87% of the mean gap remains after the name is removed.

Name is the largest single channel — not the whole gap

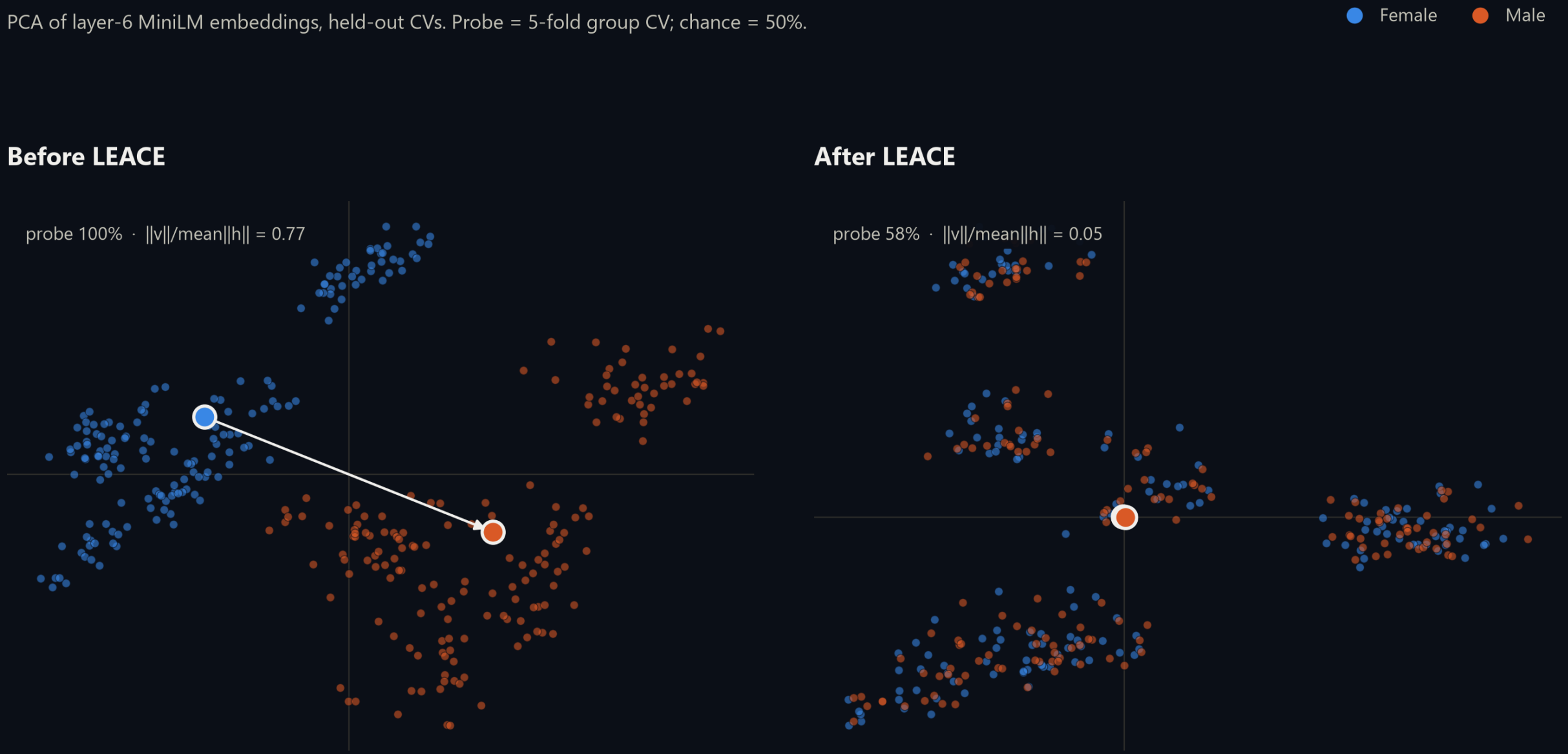
Percentage points removed from the 8.8 pp baseline gap; same ranker, held-out CVs.



Cues overlap: individual bars do not add to the combined 8.8 pp. Removing all four is the only clean cut.

The gender direction is mostly erased

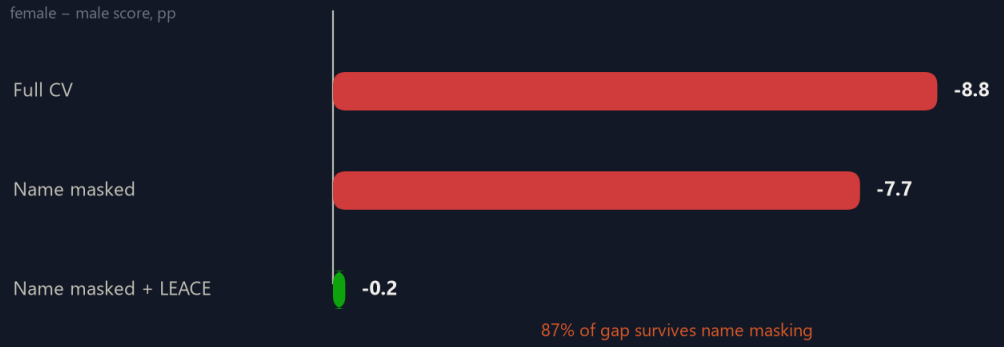
PCA of layer-6 MiniLM embeddings, held-out CVs. Probe = 5-fold group CV; chance = 50%.



LEACE is fit on the training split only, then applied to held-out representations.

BEHAVIORAL GAP

female − male score, pp



REPRESENTATION PROBE

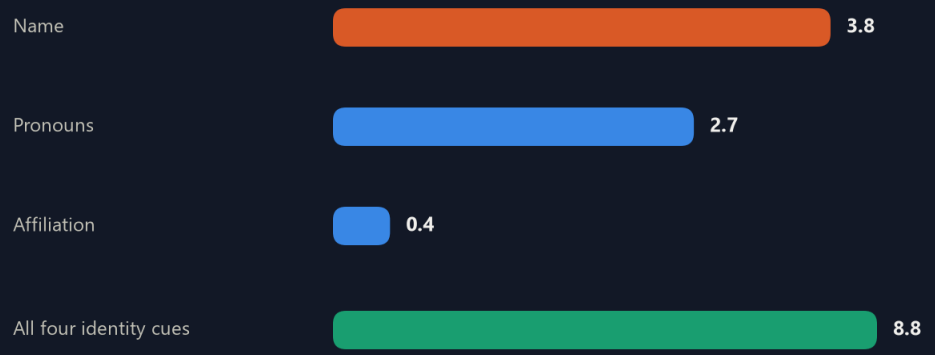
linear gender classifier, layer 6

100% → 57.8%



FEATURE CONTRIBUTION

gap removed when cue is ablated, pp



DEPLOYMENT GATE

all checks must pass together

Behavioral gap	0.2 pp
Gender probe	57.8%
Ranking AUC	0.735
Bias retention	4.7%

PASS / SHIP

Synthetic benchmark with known ground truth. Absolute scores are not a production claim.

seed 7 · 8 job families · 10 features

Scrub the representation. Gate the behavior.

A closed-form edit inside the model, plus a deployment gate that refuses to ship on one metric.

01 / SCAN

Paired audit

Find the leak channels

02 / ERASE

LEACE

Project out gender subspace

03 / RECALIBRATE

Linear head

AUC 0.742 → 0.735

04 / GATE

4 checks

Ship only if all pass

THE EDIT

$$r(h) = h - P\,h$$

P projects out the gender direction. Closed-form, deterministic, and applied to standardized layer-6 states.

WHY IT FITS AN ATS

- No retraining from scratch — only the head is recalibrated.
- Edits the representation, not the applicant.
- Cheap enough to rerun on every model update.
- Fails loudly when the probe or the gap returns.

The model changes. The applicant does not have to.

Existing fixes stop at the score. This one goes inside.

Six families of mitigation. Each leaves a different hole.

METHOD	WHAT IT FIXES	WHAT IT MISSES
Blind screening / redaction	Visible name, some fields	Proxies remain; 7.7 pp score gap here
Fairness fine-tuning (LoRA/OFT)	Output behavior on an audit set	Needs labels + compute; can over-debias (JobFair)
Adversarial debiasing	Learned representation	Unstable; hard to audit; sensitive to hyperparameters
Post-processing thresholds	Selection rates at the cutoff	Symptom only; the representation is untouched
Prompt guardrails	LLM outputs	Model-dependent; prompt injection; partial
LEACE + dual gate (ours)	Linear gender subspace + behavior	Nonlinear residual remains; needs layer choice

Behavior-only fixes act after the score. LEACE acts before it.

Better fit for the post-anonymization failure mode.

Not a universal winner — a targeted fix for the case where the name is already gone.

01 / POST-MASKING

7.5 pp

of residual gap removed: 7.7 → 0.2 pp

02 / REPRESENTATION

100 → 58%

gender probe before → after; chance = 50%

03 / RANKING QUALITY

0.742 → 0.735

AUC preserved; head recalibrated, not retrained

04 / GOVERNANCE

4 gates

behavior + probe + AUC + retention; all must pass

Blind screening removed 1.1 pp. LEACE removed 7.5 pp. The difference is where the edit happens.

Caveat: LEACE targets linear subspaces. A 58% probe means residual signal remains; the gate is the backstop.

What this proves — and what it doesn’t.

A benchmark can validate the audit. It cannot certify a production system.

WHAT IT PROVES

- The audit recovers a known injected bias.
- Name masking alone leaves most of the gap.
- LEACE reduces both behavioral and activation metrics.
- Ranking quality stays essentially flat.

WHAT IT DOES NOT PROVE

- Behavior of any specific commercial ATS.
- Real-world label bias or legal compliance.
- Race, nonbinary, disability, or intersectional effects.
- Robustness to adaptive or adversarial candidates.

NEXT 90 DAYS / RUN IT ON REAL LOGS

W1–2 / INSTRUMENT

Hook the encoder

W3–4 / AUDIT

Paired + probe baseline

W5–6 / MITIGATE

LEACE + recalibrate

W7–8 / GATE

Ship / block decision

Before any pilot: legal review, candidate notice, and an intersectional red-team.

The test suite, in one place.

Definitions used in the audit. Dots denote model outputs; A is the protected attribute.

BEHAVIORAL BIAS

$$d(P(\tilde{Y} \mid A=a), P(\tilde{Y} \mid A=a'))$$

ACTIVATION-SPACE VECTOR

$$v_l = E[h_l \mid A=a] - E[h_l \mid A=a']$$

ACTIVATION-SPACE BIAS RATIO

$$\|v_l\| / E\|h_l\|$$

DEMOGRAPHIC DISPARITY

$$\Delta = |P(\hat{Y}=1 \mid A=\text{White}) - P(\hat{Y}=1 \mid A=\text{Black})|$$

PRESENCE

$$\|v_l\| / \|\Sigma_l\|$$

RETENTION FRACTION

$$(\text{Presence}(M') - \text{Presence}(B)) / (\text{Presence}(M_{base}) - \text{Presence}(B))$$

COSINE SIMILARITY

$$\cos(\psi_{after}, \psi_{bias})$$

BALANCED ACCURACY

$$\tfrac{1}{2}(\text{qualified acceptance} + \text{unqualified rejection})$$

CV SCORE DIFFERENCE

$$d_{ij} = s(CV^f_i, job_j) - s(CV^m_i, job_j)$$

LEACE PROJECTION

$$r(h) = h - W^T(WW^T)^{-1}Wh$$

B = blind baseline; M_base = biased baseline; M′ = mitigated model; h_l = layer-l hidden state.

Sources and reproducibility.

Every statistic in this deck is traceable to the benchmark or the cited literature.

LITERATURE / 1–4

1 Wilson, K. & Caliskan, A. (2024). Gender, Race, and Intersectional Bias in Resume Screening via Language Model Retrieval. AIES. doi:10.1609/aies.v7i1.31748

2 An, J., Huang, D., Lin, C. & Tai, M. (2025). Measuring gender and racial biases in large language models. PNAS Nexus. doi:10.1093/pnasnexus/pgaf089

3 Goldin, C. & Rouse, C. (2000). Orchestrating Impartiality: The Impact of “Blind” Auditions on Female Musicians. AER 90(4), 715–741.

4 Wang, Z. et al. (2024). JobFair: A Framework for Benchmarking Gender Hiring Bias in LLMs. Findings of EMNLP. doi:10.18653/v1/2024.findings-emnlp.184

HOW THE NUMBERS MAP

8.8 / 7.7 pp = benchmark; 85.1% = (1); 1–3 pp = (2); reverse bias = (4).

METHODS / 5–8

5 Belrose, N. et al. (2023). LEACE: Perfect linear concept erasure in closed form. arXiv:2306.03819.

6 Dastin, J. (2018). Amazon scraps secret AI recruiting tool that showed bias against women. Reuters, 10 Oct.

7 Ravfogel, S. et al. (2020). Null It Out: Guarding Protected Attributes by Iterative Nullspace Projection. ACL.

8 Barocas, S. & Selbst, A. (2016). Big Data’s Disparate Impact. California Law Review 104:671.

AUDIT OUTPUTS

probe 100 → 58%; relative bias 0.77 → 0.05; retention 4.7%; AUC 0.742 → 0.735.

REPRODUCIBILITY

1,600 synthetic CVs · 800 matched pairs · 8 job families · 10 features · seed 7 · MiniLM-L6-v2, layer 6, mean-pooled states · scikit-learn 1.9.1 · concept-erasure 0.2.4 · 80/20 split by candidate · 5-fold group CV. Synthetic labels with a known injected gap; not a production audit.

Primary sources were checked against OpenAlex/DOI records; exact figures are quoted from the abstracts.